

Respiratory distress in newborn

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OUTLINES

- PULMONARY HEMORRHAGE
- PULMONARY AIR LEAKS
- PERSISTENT PULMONARY HYPERTENSION OF THE NEWBORN
- CONGENITAL DIAPHRAGMATIC HERNIA

PULMONARY HEMORRHAGE

- Occurs in about 10% of extremely preterm infants.
- **Massive** PH is less common and can be fatal.
- Autopsy demonstrates massive PH in 15% of neonates who die in the 1st 2 wk of life.
- The reported incidence at autopsy varies from 1-4 per 1,000 live births.
- Approximately 75% of affected patients weigh <2,500 g at birth.
- Prophylactic **indomethacin** in ELBW infants reduces the incidence of PH.

Why are premature babies at higher risk?

Their lungs and blood vessels are immature and more fragile

PULMONARY HEMORRHAGE....

main
RF

❖ Risk factor

- PH is usually associated with a patent ductus arteriosus.
- Rarer associations are preeclampsia, erythroblastosis fetalis, breech delivery, maternal cocaine use, coagulopathies, infection, asphyxia, hypothermia, congenital heart disease, and aspiration.

PH..PATHOGENESIS

- Pulmonary hemorrhage is, in most cases, a **hemorrhagic pulmonary edema** due to high pulmonary capillary pressures, increased filtration of fluid into the lung, and disruptions in the endothelial–epithelial barrier that cause hemorrhagic fluid to enter the air spaces.
- Underlying conditions may be **acute left ventricular failure** from hypoxia and **severe acidosis** or a **patent ductus arteriosus**.

Simple picture:

High pressure in lung capillaries



Blood/fluid leaks through damaged barrier



Blood enters alveoli



Alveoli fill with blood



Gas exchange decreases



Severe respiratory distress

Severe hypoxia = low oxygen.

Severe acidosis = blood becomes too acidic.

These can cause acute left ventricular failure

PH..PATHOGENESIS...

b/c of
Surfactant improves lung function
↓
Alveoli open better
↓
Lung resistance decreases
↓
Pulmonary blood flow can rapidly increase

- Pulmonary hemorrhage is seen with **all surfactants**;
 - Related to the **rapid increase in pulmonary blood flow** that occurs with the improvement in lung function after surfactant administration.
 - the incidence ranges from **1–5%** of treated infants and is higher with **natural** surfactant.
 - Bleeding is predominantly **alveolar** in approximately **65% of cases** and interstitial in the rest.
- **Sepsis** may be a **contributory** factor in some cases.

CLINICAL FEATURES AND DIAGNOSIS

- Increased respiratory distress, apnea, pallor, peripheral vasoconstriction, and bradycardia minutes before the onset of hemorrhage are common.
- Severe metabolic acidosis may precede the onset of hemorrhage.
- With massive hemorrhage, blood may flow out from the nose and mouth or from an endotracheal tube.
- **normal to low hematocrit.**
- **Chest radiographs** - bilateral alveolar infiltrates.

MANAGEMENT

- Immediate support and resuscitative measures
- Underlying conditions such as sepsis, asphyxia, coagulopathy, or a patent ductus arteriosus should be treated promptly.
- a high positive end expiratory pressure (PEEP) to control the bleeding
- Endotracheal administration of epinephrine to cause constriction of the pulmonary vessels
- Activated recombinant factor VII
- Exogenous surfactant therapy.
- Transfusion with packed red blood cells may be necessary to maintain adequate perfusion and blood volume.
 - However, caution should be used with excessive transfusion volumes, which can increase in left atrial pressure and pulmonary edema.

OUTCOMES

- Mortality rates following PH are high, and range from 30% to 60% in **premature** infants.
- Approximately 60% of the survivors develop chronic lung disease.
- Mortality in **term** infants is related to the underlying precipitating disease process

PULMONARY AIR LEAKS

- Air leaks may present as **extrapulmonary gas collections** in the pleural space or other locations, and can cause significant morbidity and even mortality.
- Prompt diagnosis and intervention are essential with significant air leaks.

PULMONARY AIR LEAKS...

- INCIDENCE
- **Spontaneous pneumothorax** occurs in approximately 1% of infants born vaginally at term and in approximately 2% of those born via S/C.
- **Small air leaks** in asymptomatic infants may go undetected and **resolve spontaneously**.
- More common in infants with lung disease
- Significantly **higher** in infants who require respiratory interventions at birth.
- The use of **antenatal steroids**, gentler ventilation strategies and surfactant therapy in the preterm infant have led to a decline in the incidence of air leaks in newborn infants.

Air leaks are more common in:
Babies with lung disease
Babies requiring respiratory support
Babies receiving positive-pressure ventilation

Etiology and Pathophysiology

- The most common cause of pneumothorax is **overdistention** resulting in alveolar rupture.
- **Alveolar overdistention** can occur with;
 - **Positive pressure ventilation** during neonatal resuscitation, or
 - “**ball-valve**” phenomenon that results from aspiration (classically meconium)
 - bronchial/bronchiolar obstruction.
- **Spontaneous rupture** of an underlying pulmonary malformation (e.g., lobar emphysema, congenital lung cyst, **pneumatocele**) occurs
- it is usually in an otherwise normal lung, and no etiology is identified.

Etiology and Pathophysiology...

❖ **pulmonary hypoplasia**

- **Pneumothorax** tends to occur during the **1st few hr** after birth
 - is caused by reduced alveolar surface area and poorly compliant lungs.
- It is associated with disorders of ;
 - decreased amniotic fluid volume (Potter syndrome, renal agenesis, renal dysplasia, chronic amniotic fluid leak)
 - Decreased fetal breathing movement (oligohydramnios, neuromuscular disease)
 - Pulmonary space-occupying lesions (diaphragmatic hernia, pleural effusion, chylothorax), and
 - Thoracic abnormalities (thoracic dystrophies).

Etiology and Pathophysiology...

- Air leaks often manifest with **pulmonary interstitial emphysema**.
- Air that has leaked from the alveoli may track through the perivascular and peribronchial cuffs to the hilum of the lung and rupture into the mediastinum (to produce a **pneumomediastinum**) or into the pleural space (to produce a **pneumothorax**).
- More rarely, air may track from the mediastinum to the pericardial space (producing a **pneumopericardium**) or to the peritoneal cavity (producing a **pneumoperitoneum**).

Etiology and Pathophysiology...

❖ **Tension pneumothorax occurs:**

- if an accumulation of air within the pleural space is sufficient to elevate **intrapleural pressure above atmospheric pressure**.
- Unilateral - results in **impaired ventilation** not only in the ipsilateral lung but also in the contralateral lung.
- Compression of the **vena cava** and **torsion of the great vessels** may interfere with **venous return**.

Clinical Manifestations

- **Asymptomatic** pneumothorax
 - are hyperresonance and diminished ipsilateral breath sounds with or without tachypnea.
- **Symptomatic** pneumothorax
 - is characterized by respiratory distress, which varies from merely high respiratory rate to severe dyspnea, tachypnea, and cyanosis.
 - Irritability and restlessness or apnea may be the earliest signs.
 - The onset is usually sudden but may be gradual
- **P/E:**
 - chest asymmetry with an increased anteroposterior diameter, hyperresonance, and diminished or absent breath sounds.
 - Displacement of the cardiac apex and point of maximal impulse.
 - Abdominal distention.
 - symmetry of findings does not rule it out (**bilateral in 10%**).
 - In tension pneumothorax, signs of shock are typical.

Clinical Manifestations...

❖ Pneumomediastinum:

- Can occur in patients with pneumothorax and is usually **asymptomatic**.
- The degree of respiratory distress depends on the amount of trapped gas;
 - if great, bulging of the midthoracic area is observed, ***the neck veins are distended, and blood pressure is low.***
 - are a result of **tamponade** of the systemic and pulmonary veins.
- **subcutaneous emphysema** in newborn infants is almost **pathognomonic of pneumomediastinum.**

Clinical Manifestations...

❖ Pulmonary interstitial emphysema (PIE)

- may precede the development of a pneumothorax or may occur independently
- *increasing respiratory distress* as a result of decreased compliance, hypercapnia, and **hypoxemia**.
- In severe cases, it precedes the development of BPD.
- Prevented by avoidance of high inspiratory or mean airway pressures
- **Treatment**
 - bronchoscopy - if evidence of mucous plugging
 - selective intubation and ventilation of the **uninvolved** bronchus, oxygen, general respiratory care, and HFV.

Grade 1 (Mild): Requirement for room air or nasal cannula (NC) with a flow rate of ≤ 2 L/min.

Grade 2 (Moderate): Requirement for nasal cannula flow > 2 L/min

Grade 3 (Severe): Requirement for invasive mechanical ventilation

Diagnosis

- Pneumothorax
 - Dx is established by chest radiography, with the **edge of the collapsed lung** standing out in relief against the pneumothorax.
- Pneumomediastinum
 - **hyperlucency** around the heart border and between the sternum and the heart border
- Transillumination of the thorax
- **U/S** for associated renal anomalies.
- **Pulmonary hypoplasia** is suggested by:
 - signs of uterine compression (extremity contractures)
 - a small thorax on chest radiographs
 - severe hypoxia with hypercapnia, and
 - signs of the primary disease (hypotonia, diaphragmatic hernia, Potter syndrome)

Diagnosis...

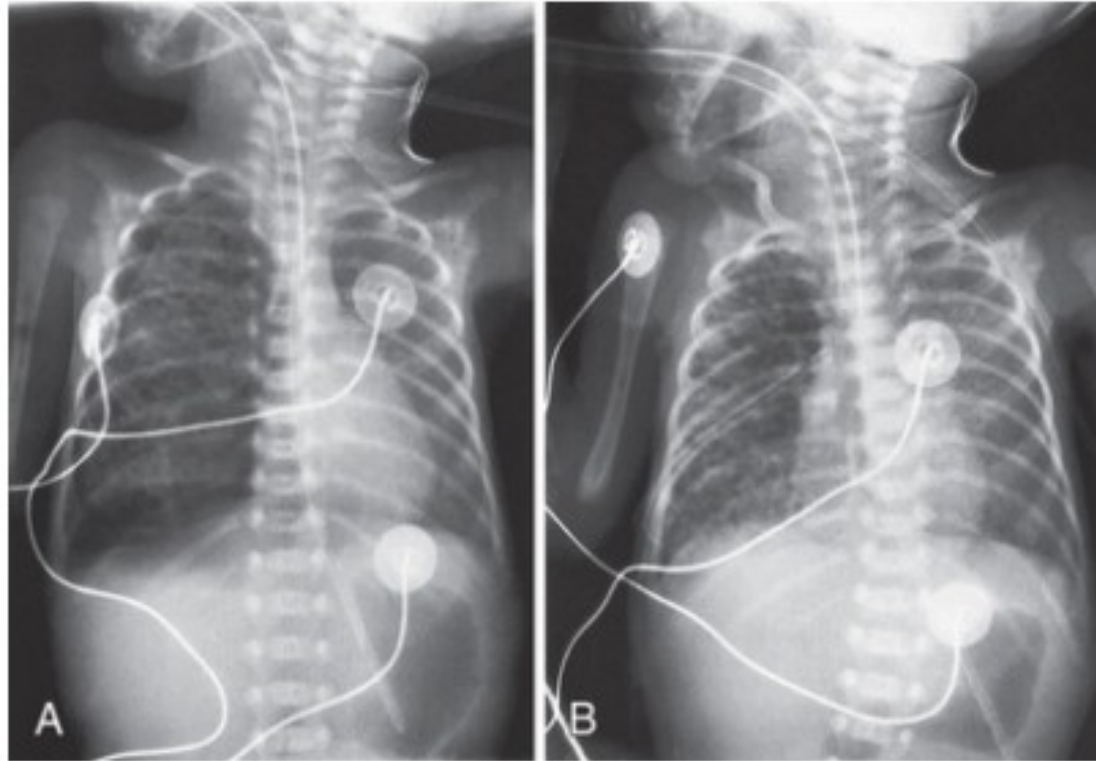


FIG. 122.13 **A**, Right-sided tension pneumothorax and widespread right lung pulmonary interstitial emphysema in a preterm infant receiving intensive care. **B**, Resolution of pneumothorax with a chest tube in place. Pulmonary interstitial emphysema (PIE) persists. (From Meerstadt PWD, Gyll C: *Manual of neonatal emergency x-ray interpretation*, Philadelphia, 1994, Saunders, p 73.)

Diagnosis...

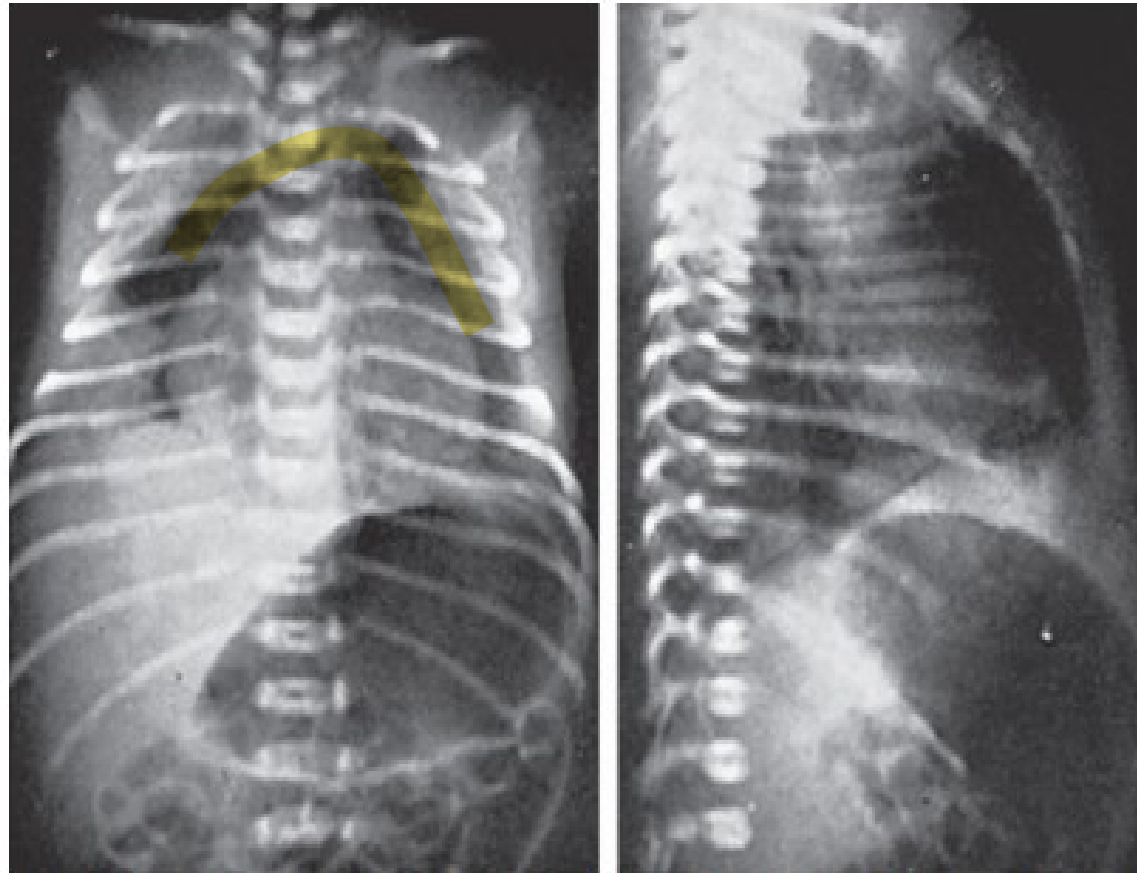


FIG. 122.14 Pneumomediastinum in newborn infant. The anteroposterior view (*left*) demonstrates compression of the lungs, and the lateral view (*right*) shows bulging of the sternum, each resulting from distention of the mediastinum by trapped air.

Diagnosis...

❑ Pneumopericardium

- may be asymptomatic, requiring only general supportive treatment, but it usually manifests as **sudden shock** with tachycardia, muffled heart sounds, and poor pulses suggesting **tamponade**.

❑ Pneumoperitoneum

- May be confused with **intestinal perforation**.
- Abdominal paracentesis.
 - **Organisms on Gram stain** suggests the intestinal perforation.
- Can result in an abdominal compartment syndrome requiring decompression.

Treatment

- Close observation – if air leak is not continuous , asymptomatic and mildly symptomatic, small pneumothoraces
 - Frequent small feedings and minimize crying
 - **Breathing 100% oxygen** in **term** infants - resorption of free pleural air into blood by reducing the nitrogen tension in blood.
- Emergency **decompression by needle thoracentesis**
- a chest tube should be inserted and attached to underwater seal drainage.
- If the air leak is **ongoing**, continuous suction (–5 to –20 cm H₂ O)
- A pneumopericardium - requires prompt evacuation of entrapped air.
- Severe localized PIE - respond to selective **bronchial intubation**.
- Surfactant therapy for RDS reduces the incidence of pneumothorax.

Persistent Pulmonary Hypertension of the Newborn

- **I. DEFINITION.** presents as severe unresponsive hypoxic respiratory failure.
 - It reflects **failure** of the transition of fetal to neonatal circulation.
 - PPHN is associated with a **high** mortality risk and risk of neurodevelopmental disabilities.
 - PPHN also has been called “persistent fetal circulation.”

PPHN...

- **II. EPIDEMIOLOGIC ASSOCIATIONS.**

- PPHN occurs at a rate of 1 to 2 per 1,000 live births and is **most common** among *full-term and post-term infants*.
- Risk factors for PPHN includes;
 - Maternal diabetes mellitus
 - Urinary tract infection during pregnancy
 - Cesarean section delivery.
 - Meconium-stained amniotic fluid
 - Maternal fever, anemia, and pulmonary disease
 - Male infants and those of black or Hispanic race are also at an increased risk for PPHN.

- **mechanisms** of antenatal pathogenesis remain uncertain
- Perinatal and neonatal conditions that have well-established links with PPHN.
- **A. Perinatal asphyxia** is the most common associated comorbidity.
 - *Long-standing fetal stress and hypoxemia- remodeling and abnormal muscularization* of pulmonary arterioles.
 - *Acute birth asphyxia* - causes release of **vasoconstricting** humoral factors and suppression of pulmonary vasodilators
- **B. Pulmonary parenchymal diseases**, RSD, pneumonia, and MAS also are associated with an increased risk of PPHN.
 - In most such cases, the PHN is **easily reversible**, suggesting a **transient** vasospasm.
 - The risk of PHN is **greater** when the fetus is of more **advanced** gestational age.

- **C.** Abnormalities of *pulmonary development* contribute structurally to **PPHN**, either by
 - **Pruning** of the vascular tree
 - congenital diaphragmatic hernia, Potter's syndrome, and other forms of pulmonary parenchymal hypoplasia, **or**
 - **Malalignment** of pulmonary veins and arteries
 - alveolar capillary dysplasia.
- **D.** Pneumonia and/or sepsis of bacterial or viral origin can initiate PPHN.
 - pathophysiologic mechanisms
 - suppression of endogenous nitric oxide (NO) production
 - endotoxin-mediated myocardial depression, and
 - pulmonary vasoconstriction associated with release of thromboxanes.

- **III. PATHOLOGY AND PATHOPHYSIOLOGY**

- **A. Pulmonary vascular remodeling**

- is **pathognomonic** of **idiopathic PPHN** and **fatal PPHN**.
- **Abnormal muscularization** of the normally nonmuscular intra-acinar arteries, with increased medial thickness of the larger muscular arteries-
 - results in a decreased cross-sectional area of the pulmonary vascular bed and elevated PVR.
- One possible stimulus to pulmonary vascular remodeling is **fetal hypoxemia**.

- **.B. Pulmonary hypoplasia** - affects both alveolar and pulmonary arteriolar development.

- It may be an isolated anomaly or be a part of congenital diaphragmatic hernia, oligohydramnios syndrome, or renal agenesis (i.e., Potter's syndrome)

- **C. Reversible pulmonary vasospasm, nonfatal PPHN**
 - Is often due to lung disease, infections, or acute asphyxia.
 - Hypoxia induces profound pulmonary VC, and this response is exaggerated by acidemia.
 - **Suppression** of endogenous NO, prostacyclin, or bradykinin production and release of thromboxanes (A2 and its metabolite, B2), and leukotrienes (C4 and D4)
 - appear to mediate the increased PVR seen with **sepsis and hypoxemia**.

- **D. Myocardial dysfunction with elevated PVR.**
- Congenital diaphragmatic hernia is often associated with intractable PPHN.
- A combination of RV **hypertrophy** and/or failure and LV **hypoplasia** (because of reduced pulmonary venous return during fetal life) with pulmonary venous hypertension results in severe PPHN.
 - will **not** respond to conventional pulmonary vasodilator therapy, until ventricular function improves.

- **1. RV dysfunction** can be caused by intrauterine **constriction of the ductus arteriosus**, which results in higher PVR, RV failure, and an **atrial** right-to-left shunt.
 - Furthermore, RV failure resulting in altered **diastolic** compliance causes right-to-left atrial shunting, even in the absence of elevated PVR.
- **2. LV dysfunction** causes pulmonary venous hypertension and secondary pulmonary arterial hypertension (PAH), contributing to **right-to-left hemodynamic shunting through the ductus arteriosus**.
 - Treating this form of pulmonary hypertension requires an approach that improves LV function rather than simply lowering PVR.

- **E. Mechanical factors** that influence PVR include cardiac output and blood viscosity.
 - Hyperviscosity(polycythemia), reduces pulmonary microvasculature perfusion.

- **IV. DIAGNOSIS.** PPHN should always be considered in evaluating a **cyanotic newborn**
- **A. DDX OF PPHN** - CCHD, sepsis, and severe pulmonary parenchymal disease.
- **B.** The infant with PPHN appears **very sick** and has **unresponsive** cyanosis
- **C.** A gradient of **10% or more in oxygenation saturation** between simultaneous preductal (right upper extremity) and postductal (lower extremity) arterial blood gas (ABG) values
- Because a subset of infants with PPHN has **closure of the ductus arteriosus** and their hemodynamic shunting occurs only at the foramen ovale, **the absence of differential cyanosis or SaO2 does not exclude PPHN.**

- **D.** The **chest radiograph** usually appears **normal** or shows associated pulmonary parenchymal disease.
 - The **cardiothymic silhouette is normal**, and **pulmonary blood flow is normal** or **diminished**.
- **E.** The **electrocardiogram (ECG)**
- Most commonly shows **RV predominance** that is within the range considered normal for age.
- Less commonly, an axis that is left (or “normal”) suggests poor RV function and poor prognosis.
- An **abnormal axis** may also point to **CCHD**.

- **V. MANAGEMENT.**
- PPHN is a **medical emergency** in which immediate, appropriate intervention is critical to reverse hypoxemia, improve pulmonary and systemic perfusion, and preserve end-organ function.
- Adequate respiratory support to optimize oxygenation, ventilation (carbon dioxide), and acid–base status (pH) facilitates the fall in PVR.

CONGENITAL DIAPHRAGMATIC HERNIA

❖ Epidemiology

- The incidence of CDH is between 1 in 2,000 and 1 in 5,000 live births.
- Females affected twice as often as males.
- More common on the **left** (85%) and are occasionally **bilateral** (<5%).
- Pulmonary hypoplasia and malrotation of the intestine are part of the lesion, **not associated anomalies**.

CDH...

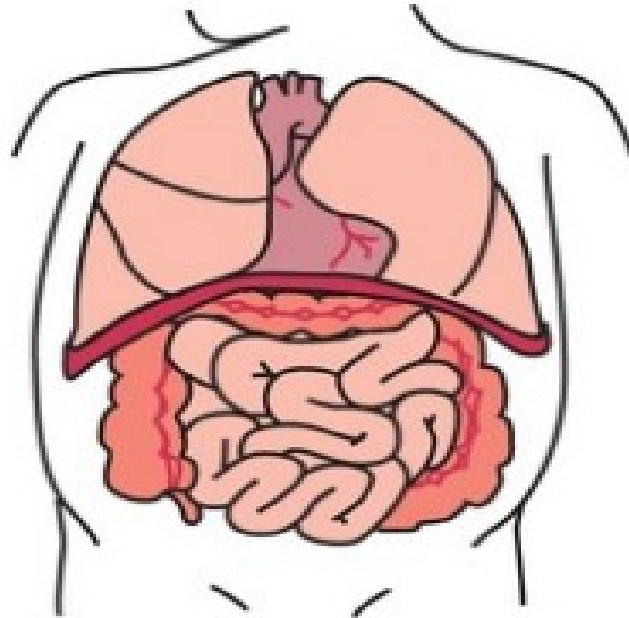
- Most cases of CDH are **sporadic**, but familial cases have been reported.
- Associated **anomalies** have been reported in up to 30% of cases, including CNS lesions, esophageal atresia, omphalocele, and cardiovascular lesions.
- CDH is recognized as part of several **chromosomal** syndromes: trisomies 21, 13, and 18 and Fryns, Brachmann–de Lange, Pallister-Killian, and Turner syndromes.

CDH...

❖ TYPES OF DIAPHRAGMATIC HERNIA

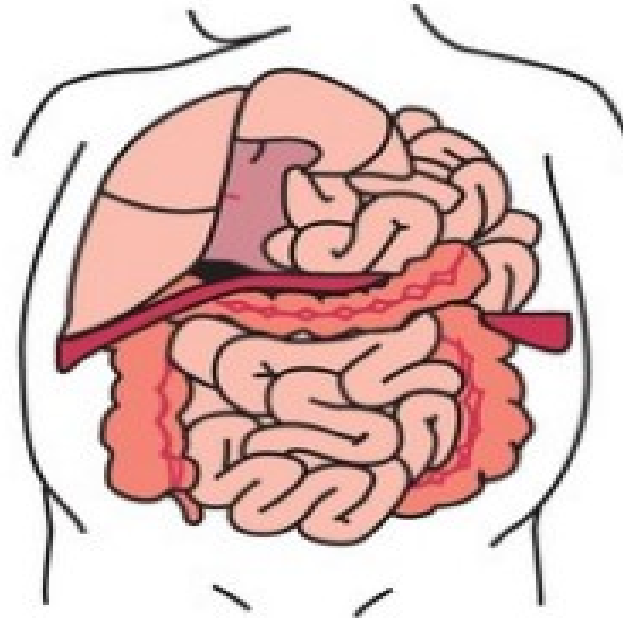
- a Bochdalek hernia (most common type of CDH), occurs through a **posterior** defect in the diaphragm.
- a Morgagni hernia, anterior retrosternal hernia, the central and lateral portions of the diaphragm
- the stomach, spleen, and most of the intestines herniate into the thorax.
- In rare instances, the liver and kidneys may also be in the thoracic cavity.
- Herniation of the abdominal content- results in pulmonary hypoplasia that is most severe on the ipsilateral side.
- Pulmonary **capillary blood flow** is decreased.
- **Surfactant production** may also be affected by pulmonary hypoplasia in infants with CDH.

CDH...



Normal diaphragm

A



Bochdalek diaphragmatic defect
with herniation of small lung

B

FIG. 122.10 **A**, A normal diaphragm separating the abdominal and thoracic cavity. **B**, Diaphragmatic hernia with a small lung and abdominal contents in the thoracic cavity.

CDH...

❖ PRENATAL DIAGNOSIS

- In >50% of cases, CDH can be diagnosed on prenatal US between 16 and 24 wk of gestation.
- High-speed fetal MRI can further define the lesion.
- **Fetal ultrasound:-**
 - reveal abdominal organs and fluid-filled bowel with peristalsis in the thorax, and
 - a shift of the heart and mediastinum away from the side of the hernia.
- Polyhydramnios, pleural effusions, and ascites often are present.
- Fetal Echo. and karyotype – for diagnosis of common associated malformations.

CDH...

□ CLINICAL FEATURES

- Depends on the type and size of the hernia.
 - ❖ **large left-sided hernia**
 - may present with a scaphoid abdomen and significant respiratory symptoms in the **delivery room**.
 - Breath sounds may not be heard on either side of the chest
 - bowel sounds may be audible in the chest, and
 - the heart sounds may be better heard on the right side.
 - The infant's respiratory status may worsen as air is swallowed and fills the bowel in the chest, compressing the lungs.
 - Chest and abdominal radiographs should be done after an **orogastric tube** is placed.

❖ Right-sided hernia

- may be *asymptomatic* at birth, and the diagnosis may be incidental if not made prenatally.
- the **liver** and some intestine may occupy the right hemithorax, but little or no displacement of the heart may occur.
- have a **better prognosis** because they generally have less lung hypoplasia.



FIGURE 52-5 Left-sided congenital diaphragmatic hernia. An anteroposterior radiograph of a near-term infant with known left diaphragmatic hernia, intubated in delivery room. Note the bowel in the left chest, with displacement of the heart to the right. Incidental finding on the radiograph was the orogastric tube noted high in the esophagus. This patient also had esophageal atresia.

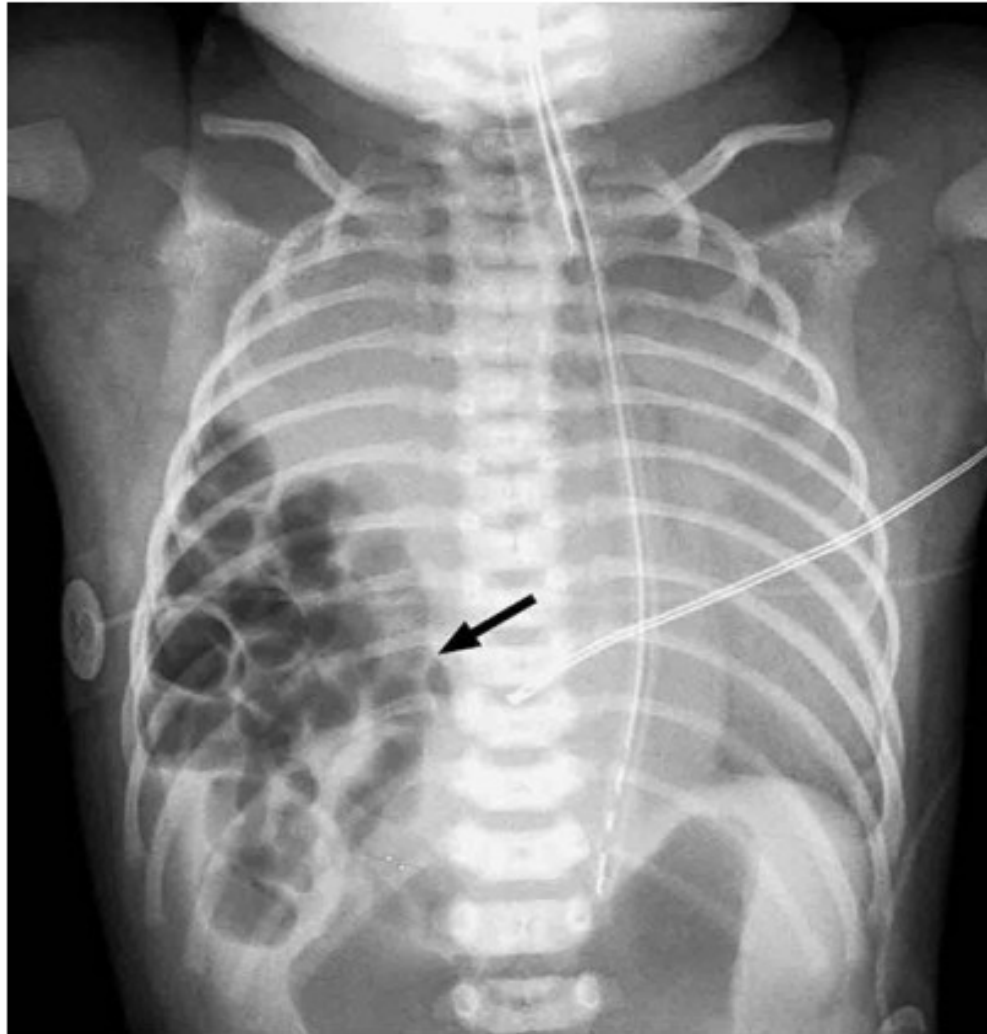


FIGURE 52-6 Right-sided congenital diaphragmatic hernia. An anteroposterior chest radiograph reveals gas-distended bowel loops (*arrow*) projecting over the liver shadow as well as right lower lung fields. There is opacification of the bilateral lung fields and little or no displacement of the heart.

- Approximately 40% of infants with CDH have associated anomalies
- During fetal life, the left ventricle is often small (probably as a result of increased intrathoracic pressure) and left ventricular output is reduced.
- Postnatally, the left ventricular size and output may improve with correction of the hernia.

❑ MANAGEMENT

- a **double-lumen orogastric tube** should be placed into the infant's stomach in the delivery room to reduce air in the bowel and decrease compression of the lung.
 - **If assisted ventilation** (via an endotracheal tube)
 - Bag and mask ventilation should be **avoided**.
 - Close monitoring of oxygenation and radiological evaluation
 - umbilical catheters should be placed for medications and for blood gas monitoring.
 - Sedation may be needed to avoid the need for higher pressures and occurrence of pneumothorax.
- POSTNATAL CARE
 - Directed toward lung function and pulmonary hypertension.
 - Surgical intervention is delayed where possible, until the infant is hemodynamically stable and pulmonary hypertension is controlled.
 - Mechanical ventilation is currently based on principles of **“gentle ventilation.”**

- MEDICAL INTERVENTIONS
- Surfactant
 - Surfactant therapy in CDH has **no reported benefits**, and may cause clinical deterioration and increased mortality.
- Ventilation Strategy
 - **a gentler ventilation strategy**, which allows for higher PaCO₂ and lower preductal oxygen saturation as long as blood pressure and tissue perfusion are maintained.

- Nitric Oxide
 - it has not had the same success for infants with CDH like PPHN.
- Extracorporeal Membrane Oxygenation ECMO
 - is now used more frequently during preoperative stabilization.

❖ SURGICAL REPAIR

- Surgical repair of the diaphragmatic defect should be performed during the **neonatal period**.
- Fetal surgical procedures for CDH have **failed** to improve survival
- **Currently**, the preferred management strategy for CDH is medical stabilization with appropriate cardiorespiratory support.

- **OUTCOMES**
- Position of the liver and the lung area-to-head circumference ratio (LHR) via ultrasound is important.
 - There were no survivors with LHR <1, but all babies with LHR >1.4 survived.
 - liver in the thoracic cavity is a poor prognostic feature.
- Overall survival of live born infants with CDH is 71%.
- Relative predictors of a poor prognosis include:
 - an associated major anomaly
 - symptoms before 24 hr of age
 - severe pulmonary hypoplasia
 - herniation to the contralateral lung, and
 - the need for ECMO.
- The size of the defect appears to be the strongest predictor of morbidity.

- Survivors are at risk for chronic lung disease, feeding difficulties, gastroesophageal reflux, scoliosis, pectus excavatum, hearing loss, neurodevelopmental delay, brain injury, and recurrence of the diaphragmatic hernia.
- Overall survival of patients with CDH and significant cardiac defects is lower than those without heart abnormalities.

THANK YOU